

Problem Set 2 – Set identities

Session 2 · Sets & set operations

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

SOLUTION (a) $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$.
 (b) $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$; $A \setminus B = \{2, 4, 8, 10\}$;
 $A \triangle B = \{2, 3, 4, 8, 9, 10\}$; $B^c = \{1, 2, 4, 5, 7, 8, 10, 11\}$.

2 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$
- $\{1\} \in C$
- $\{1\} \subseteq C$
- $\{1, 2\} \subseteq C$
- $\emptyset \subseteq C$
- $|C| = 4$

SOLUTION (a) True – 1 is listed. (b) True – $\{1\}$ is listed as an element. (c) True – the only element of $\{1\}$ is 1 , and $1 \in C$. (d) False – it would need $2 \in C$, and 2 is not an element of C . (e) True – vacuously, for every set. (f) True.

3 POWER SET

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for *any* finite A .

SOLUTION (a) $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.

(b) $|\mathcal{P}(A)| = 8 = 2^3$. Building a subset means deciding, independently for each of the n elements, whether it is in or out – two choices n times, so 2^n subsets.

4 SET-BUILDER

Let $S = \{x \in \mathbb{Z} : x^2 < 10\}$.

- Write S in roster notation, and say in one sentence how you know the list stops where it does.
- Write $\{2, 4, 6, 8\}$ in set-builder notation.
- Is $-3 \in S$? Is $\{0, 1\} \subseteq S$? Justify each from the definition.

SOLUTION (a) $S = \{-3, -2, -1, 0, 1, 2, 3\}$. For an integer, $x^2 < 10$ says $|x| \leq 3$, since $3^2 = 9 < 10$ but $4^2 = 16 > 10$ – and the condition depends only on $|x|$, so the list is symmetric about 0.

(b) $\{x \in \mathbb{Z} : x \text{ is even and } 0 < x < 10\}$, or equivalently $\{2k : k \in \mathbb{Z}, 1 \leq k \leq 4\}$.

(c) Yes, $-3 \in S$: $(-3)^2 = 9 < 10$, so -3 satisfies the defining condition. Yes, $\{0, 1\} \subseteq S$: every element of $\{0, 1\}$ – namely 0 and 1 – lies in S , which is exactly what $\subseteq$ demands.

5 THE EMPTY SET

Decide true or false, and in each case name the definition that settles it.

- $\emptyset \in \{\emptyset\}$
- $\emptyset \subseteq \{\emptyset\}$
- $|\{\emptyset\}| = 1$
- $\emptyset = \{\emptyset\}$
- $\mathcal{P}(\emptyset) = \{\emptyset\}$

SOLUTION (a) True – $\{\emptyset\}$ has exactly one element listed, and that element is $\emptyset$. (b) True – $\emptyset \subseteq X$ for every set X , vacuously: there is no element of $\emptyset$ that could fail to be in X . (c) True – one element, namely $\emptyset$. (d) False – $\emptyset$ has no elements and $\{\emptyset\}$ has one, so by extensionality they are different sets. An empty box inside a box is not the same as an empty box. (e) True – the only subset of $\emptyset$ is $\emptyset$ itself, and $2^0 = 1$ agrees.

Part B · Six identities to prove

6 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \cup (B \cap C)$ be arbitrary. Then $x \in A$, or $x \in B \cap C$. If $x \in A$ then x is in both $A \cup B$ and $A \cup C$, so in their intersection. If $x \in B \cap C$ then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$. Either way $x \in (A \cup B) \cap (A \cup C)$. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$ be arbitrary, so $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$ we are done. If not, then from the first $x \in B$ and from the second $x \in C$, so $x \in B \cap C$. Either way $x \in A \cup (B \cap C)$. ■

7 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

SOLUTION ($\subseteq$) Let $x \in (A \cap B)^c$ be arbitrary. Then $x \in U$ and $x \notin A \cap B$, i.e. it is not the case that both $x \in A$ and $x \in B$. So $x \notin A$ or $x \notin B$, giving $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$. ($\supseteq$) Let $x \in A^c \cup B^c$ be arbitrary. Then $x \notin A$ or $x \notin B$, so it is not the case that x lies in both, i.e. $x \notin A \cap B$. As $x \in U$, $x \in (A \cap B)^c$. ■

8 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \setminus (B \cup C)$ be arbitrary. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$, so x is in the intersection. ($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. Then $x \in A$, $x \notin B$ and $x \notin C$, so $x \notin B \cup C$, giving $x \in A \setminus (B \cup C)$. ■

9 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

SOLUTION Recall $A \triangle B = (A \setminus B) \cup (B \setminus A)$. ($\subseteq$) Let $x \in A \triangle B$ be arbitrary, so $x \in A \setminus B$ or $x \in B \setminus A$. In the first case $x \in A \subseteq A \cup B$ and $x \notin B$, so $x \notin A \cap B$. The second case is symmetric. Either way $x \in (A \cup B) \setminus (A \cap B)$. ($\supseteq$) Let $x \in (A \cup B) \setminus (A \cap B)$ be arbitrary. Then x is in A or in B , but not in both. If $x \in A$ then $x \notin B$, so $x \in A \setminus B$; otherwise $x \in B$ and $x \notin A$, so $x \in B \setminus A$. Either way $x \in A \triangle B$. ■

10 SUBSETS

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

SOLUTION $(\Rightarrow)$ Suppose $A \subseteq B$. Certainly $B \subseteq A \cup B$. Conversely, let $x \in A \cup B$ be arbitrary: if $x \in B$ we are done, and if $x \in A$ then $x \in B$ by the hypothesis. So $A \cup B \subseteq B$, and the two sets are equal.
 $(\Leftarrow)$ Suppose $A \cup B = B$. Let $x \in A$ be arbitrary. Then $x \in A \cup B$, which equals B , so $x \in B$. Hence $A \subseteq B$. ■

11 CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

SOLUTION Elements here are ordered pairs, so an arbitrary element is written (x, y) .
 $(\subseteq)$ Let $(x, y) \in A \times (B \cap C)$ be arbitrary. Then $x \in A$ and $y \in B \cap C$, so $y \in B$ and $y \in C$. Hence $(x, y) \in A \times B$ and $(x, y) \in A \times C$.
 $(\supseteq)$ Let $(x, y) \in (A \times B) \cap (A \times C)$ be arbitrary. Then $x \in A, y \in B$ and $y \in C$, so $y \in B \cap C$ and $(x, y) \in A \times (B \cap C)$. ■

Part C · Counterexamples and writing

12 DISPROVE

Each claim below is false in general. For each one: name explicit finite sets, compute **both** sides, and point at an element that is in one side and not the other.

- $A \setminus (B \setminus C) = (A \setminus B) \setminus C$
- $A \times B = B \times A$
- $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$

SOLUTION (a) Take $A = B = C = \{1\}$. Then $B \setminus C = \emptyset$, so the left side is $A \setminus \emptyset = \{1\}$; and $A \setminus B = \emptyset$, so the right side is $\emptyset$. The element **1** is in the left side and not the right.
 (b) Take $A = \{1\}, B = \{2\}$. Then $A \times B = \{(1, 2)\}$ and $B \times A = \{(2, 1)\}$, and $(1, 2) \neq (2, 1)$ because ordered pairs are equal only when they agree coordinate by coordinate. So $(1, 2)$ is in the left side and not the right. (Any counterexample needs $A \neq B$ and both non-empty – if either is empty both products are $\emptyset$.)
 (c) Take $A = \{1\}, B = \{2\}$. Then $\{1, 2\} \subseteq A \cup B$, so $\{1, 2\} \in \mathcal{P}(A \cup B)$. But $\mathcal{P}(A) = \{\emptyset, \{1\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{2\}\}$, whose union contains only $\emptyset, \{1\}, \{2\}$ – not $\{1, 2\}$. Note the $\cap$ version *is* true: a set is a subset of both A and B exactly when it is a subset of $A \cap B$.

A student hands in the following.

Claim. $A \cap (B \cup C) = (A \cap B) \cup C$ for all sets A, B, C .

Proof. Draw three overlapping circles. Shade $A \cap (B \cup C)$, then shade $(A \cap B) \cup C$. Both shadings cover the region in the middle where all three circles meet, so the two sets are equal. ■

- Which step of this argument fails, and why? Answer in one or two sentences.
- Give a counterexample to the claim.

SOLUTION

(a) The step that fails is the conclusion: agreeing on *one* region does not make two sets equal — set equality requires that they agree on **every** element, so every region would have to match. If C contains a point outside A , that point is shaded on the right and not on the left.

(b) Take $A = \emptyset$, $B = \emptyset$, $C = \{1\}$. Then $A \cap (B \cup C) = \emptyset \cap \{1\} = \emptyset$, while $(A \cap B) \cup C = \emptyset \cup \{1\} = \{1\}$. The element 1 is in the right side and not the left, so the claim is false.